FRONTEND DEVELOPMENT OF BAYUDATA: THE OFFICIAL DATA PORTAL FOR THE SABAH STATE GOVERNMENT

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ABSTRACT

Open data plays a vital role in modern governance by enhancing transparency, efficiency, and innovation. Despite its importance, the Sabah state government currently lacks a centralized platform to manage and share government data, limiting accessibility for agencies and the public. This project focuses on designing the frontend for "BayuData" Sabah's official data portal, developed in collaboration with the Bahagian Data Raya & Perkongsian Maklumat (BDRPM) team under Jabatan Perkhidmatan Komputer Negeri (JPKN).

The frontend interface was created using HTML, CSS, and JavaScript to ensure it is both responsive and accessible. To determine essential features and finalize the design, existing data portals from various countries were reviewed for inspiration. Feedback from the BDRPM team and the industry supervisor was also incorporated throughout the development process to refine both functionality and usability.

The final design includes a user-friendly landing page with straightforward navigation through a menu bar, filters for dataset browsing, a dataset details page with preview functionality, and error pages (404 and 403) to help users easily return to the main menu. With its responsive and intuitive interface, the prototype lays the groundwork for backend integration and further development, facilitating Sabah's efforts toward open data adoption and accessibility.

Keywords: Open Data, Open Government Data, Data Portal, Frontend Development, BayuData, Responsive Design, HTML, CSS, Government Transparency

1. INTRODUCTION

The Jabatan Perkhidmatan Komputer Negeri (JPKN), established as Unit Perkhidmatan Komputer (UPK) in 1976 and officially recognized under its current name in 1992, serves as the central IT service provider for the Sabah State Public Sector. Over the decades, JPKN has played a pivotal role in advancing IT adoption across government agencies. Its core services include data processing, infrastructure management, software development, training, and support.

The Bahagian Data Raya & Perkongsian Maklumat (BDRPM) is one of several specialized units under JPKN. This unit focuses on harnessing the power of data analytics and fostering collaborative information sharing across government departments. Its primary mission is to empower decision-making through data-driven insights, enhance operational efficiency, and promote transparency and accountability within the Sabah State Public Sector.

This project involves the development of "BayuData", Sabah's official open data portal. BayuData is designed to serve as a centralized platform for Sabah State Government agencies, private sectors, and citizens, providing access to a wide range of data for purposes such as analysis, learning, and innovation. The project was undertaken in collaboration with the BDRPM team under JPKN during the industrial training period.

Open data is a powerful enabler for transparency, accountability, and innovation. It allows unrestricted access to datasets that can be freely shared, reused, and analysed by citizens, government agencies, and private sectors for various purposes. The BayuData portal aims to address the current lack of an official open data platform in Sabah, providing a centralized and reliable source for state-wide datasets. By empowering public access to government information, BayuData supports informed decision-making, social and economic problem-solving, and the development of innovative applications and services. It emphasizes ease of access, data-driven insights, and the democratization of information. BayuData's implementation will ultimately benefit the public, researchers, and organizations, fostering a data-driven culture within the Sabah State Government.

The objectives of this project are:

1. To research and refer to existing data portals to identify and determine features to be implemented in BayuData.
2. To design and develop the frontend of BayuData, ensuring a user-friendly and responsive interface.

The scope of this project focuses on the frontend development of BayuData, Sabah's official open data portal, emphasizing key aspects of user interface (UI) design and functionality. The project includes:

- UI Design
  - Creating an intuitive and visually appealing user interface to ensure ease of navigation and engagement for users of varying technical expertise.
- Responsive Web Design
  - Ensuring the website adapts seamlessly to different screen sizes, including desktops, tablets, and mobile devices.
- Feature Implementation
  - Developing and integrating essential features such as dataset browsing, dataset preview functionality, and dataset download options to meet user needs.
- Cross-Browser Compatibility
  - Testing the portal across multiple browsers (e.g., Google Chrome, Mozilla Firefox, and Opera GX) on a local host to ensure consistent performance and appearance.
- Testing and Debugging
  - Conducting various tests to ensure functionality and usability, including:
    - Responsiveness Testing: Verifying the website's adaptability to different screen resolutions.
    - Usability Testing: Assessing the user-friendliness of navigation, layout, and feature access.

- Functional Testing: Ensuring features like dataset preview and download buttons operate as intended.

- Static Data Loading: Loading a single dataset from the database as a sample to validate the functionality of dataset-related features, serving as a prototype for future data integration.

2. RELATED WORKS

In developing BayuData, analysing existing open data portals is crucial to identify best practices that enhance usability and functionality. This section examines five government open data platforms—those of Malaysia, the UK, the US, Singapore, and the European Union—by comparing them based on navigation, general performance, data findability, understandability, and quality. These criteria are derived from the integrated usability framework proposed by Molodtsov and Nikiforova (2024), which emphasizes accommodating user diversity, facilitating collaboration, and enhancing data exploration. By evaluating these aspects, this study provides insights to guide the frontend development of BayuData.

Navigation

Effective navigation is crucial for ensuring users can access relevant datasets without frustration. According to the Nielsen Norman Group, users who become lost or stuck are more likely to abandon the task or leave for a competitor's website, reducing engagement and accessibility (Nielson Norman Group, 2016). Their usability heuristics also highlight the importance of User Control and Freedom, where clearly marked "emergency exits" allow users to undo actions or return to familiar pages, reducing frustration and improving navigation (Nielson Norman Group, 2024). While some of Nielson Norman group articles are several years old, their findings remain relevant, as intuitive navigation is consistently linked to better engagement and discoverability in digital platforms.

A user-friendly menu design, clear error handling, and consistent navigation across devices are key elements that contribute to a seamless experience. The table below evaluates selected open data portals based on these criteria.

Table 2.1 Navigation Comparison of Malaysia, UK, US, Singapore, and EU Open Data Portals

Country	User-Friendly Menu	Error Handling & Fallbacks	Consistent Navigation Across Devices
Malaysia	Aesthetically pleasing	No explicit homepage button	Inconsistent padding in mobile error display
United Kingdom	Simple and Functional	Clear error handling with homepage link	Mobile menu appears cramped in a 3-column grid layout
United States	Landing page leads well to data categories	No clear homepage return option	Tightly packed links may frustrate some users
Singapore	Aesthetically pleasing	Only shows a 404 error with no guidance for returning to homepage	Consistent design across desktop and mobile

European Union	Clean design but carousel placement may disrupt scrolling	Error page includes a clear homepage link	Carousel position may frustrate users trying to scroll down

The comparison reveals that while all five portals feature user-friendly menu designs, several face challenges with error handling and mobile consistency. The UK and EU portals demonstrate the strongest balance, offering clear homepage return links and effective error handling. However, both have room for improvement in their mobile layouts — the UK's cramped 3-column design and the EU's poorly placed carousel disrupt navigation flow.

Malaysia and the US portals lack clear homepage return options on error pages, posing a risk of users feeling stranded. The Singapore portal offers an aesthetically pleasing menu but suffers from the weakest error-handling experience, with no visible guidance for returning to the homepage.

General Performance

General performance is crucial for ensuring a smooth and efficient user experience. According to a research made in 2024, it was highlighted that SEO techniques, such as optimizing website structure and improving loading speeds, significantly enhance user engagement (Bansal, 2024). Similarly, Strikepoint Media emphasized in their article that websites loading within three seconds are more likely to retain visitors, while slower sites risk higher bounce rates and reduced satisfaction (Strikepoint, 2023).

By maintaining fast load times, stable functionality, and compatibility across devices and browsers, open data portals can provide a seamless experience that encourages user engagement. Evaluating these factors helps identify strengths and areas for improvement to guide BayuData's frontend development.

The following table compares the general performance of Malaysia, UK, US, Singapore, and EU open data portals based on key performance criteria.

Table 2.2 General Performance Comparison of Malaysia, UK, US, Singapore and EU Open Data Portals

Country	Responsive Design	Absence of Critical Errors	Fast Page Load Times	Browser Compatibility
Malaysia	Inconsistent padding in mobile view	No major errors	Fastest load time	Functions well on major browsers
United Kingdom	Cramped category layout in mobile view	No major errors	Acceptable load time	Functions well on major browsers
United States	Dataset list requires scrolling past a long list of filters on mobile	No major errors	Acceptable load time	Functions well on major browsers
Singapore	Responsive across devices	No major errors	Acceptable load time	Functions well on major browsers

European Union	Carousel placement may disrupt scrolling experience on smaller screens	No major errors	Acceptable load time	Functions well on major browsers

In terms of performance, the Malaysia portal stands out for its notably fast load times, though it faces minor responsiveness issues in mobile layouts. The Singapore portal performs well across all criteria, demonstrating strong responsiveness, speed, and compatibility.

The UK and US portals present minor usability concerns, particularly in mobile views — the UK's cramped layout and the US's filter-heavy design create unnecessary friction for users. Meanwhile, the EU portal's carousel placement may disrupt scrolling for users on smaller screens, posing a risk to smooth interaction.

Data Understandability

Effective data presentation and clear metadata are essential for improving data understandability. According to Hassenstein and Vanella (2022), clear and detailed metadata significantly enhances a user's ability to comprehend datasets by providing accurate descriptions, structure, and context. They emphasize that metadata completeness directly correlates with improved dataset interpretation, reducing confusion and enabling users to assess the data's relevance and utility. Ensuring datasets are well-documented with comprehensive metadata can greatly improve user engagement and overall platform usability.

To evaluate how various portals support data understandability, the table below compares the dataset presentation, metadata clarity, and dataset reuse count across selected government open data portals.

Table 2.3 Data Understandability Comparison of Malaysia, UK, US, Singapore and EU Open Data Portals

Country	Dataset Presentation and Organization	Metadata Clarity	Dataset reuse count
Malaysia	Sorted by category; previews available (limited data records)	Clear and concise descriptions; comprehensive detailed metadata	Download count displayed
United Kingdom	Sorted by best match or most recent; no previews	Clear descriptions; lacks variable definitions	Not displayed
United States	Sorted by popularity by default; no previews	Clear descriptions; lacks variable definitions	View count displayed
Singapore	Sorted by relevance or last updated; previews available	Some datasets lack descriptions or detailed variable definitions	Highlights popular datasets in the menu

European Union	Sorted by relevance or date modified; limited previews	Incomplete metadata for some datasets; almost no license info displayed	Not displayed

The comparison reveals key differences in how each portal presents and organizes datasets, provides metadata clarity, and displays dataset reuse counts.

Malaysia demonstrates a well-structured approach, offering clear descriptions, comprehensive metadata, and a visible download count. The US portal follows a similar structure but lacks dataset previews and variable definitions, which may hinder users seeking detailed insights. The UK portal maintains clarity in descriptions but similarly omits dataset previews and detailed variable definitions.

The EU portal presents some concerns, particularly incomplete metadata and missing license information for most of their datasets, which may affect data reliability. Meanwhile, Singapore stands out for its visually appealing dataset previews and popular dataset highlights, though some metadata inconsistencies are present.

Overall, Malaysia's balanced approach combining clear metadata, dataset previews, and a visible download count aligns well with best practices in data understandability. Singapore's strategy of featuring popular datasets in its menu is also notable as an effective method to draw user attention, even if reuse counts are not integrated directly into the metadata.

Data Quality

While clear dataset presentation and metadata clarity are crucial for helping users quickly grasp a dataset's content, its completeness and accuracy are equally important indicators of overall data quality. Incomplete or inconsistent metadata, as observed in some portals, may suggest potential issues with data reliability, timeliness, or structure — all of which can hinder effective data use. According to Timmerman et al. (2023), data quality issues can significantly impact task performance, with improvements reducing the time required to complete tasks by up to 65%. This highlights the importance of clear and comprehensive metadata to improve data understandability and ensure users can efficiently interpret dataset content.

To better assess these aspects, the following table compares key data quality elements across the examined portals, including machine-readable formats, metadata completeness, dataset download functionality, and version control practices.

Table 2.4 Data Quality Comparison of Malaysia, UK, US, Singapore, and EU Open Data Portals

Country	Machine-readable formats	Metadata Completeness	Dataset Download Functionality	Version Control for Datasets
Malaysia	CSV and Parquet available for all datasets	Complete and detailed metadata	Available	Older data compiled into one dataset; unclear if version control is implemented

United Kingdom	Various formats; CSV/XML uncommon	Mostly complete; lacks variable definitions	Available	Older versions available with upload dates clearly labelled
United States	Various formats; CSV/XLS commonly available	Mostly complete; lacks variable definitions	Available	Older data compiled into one dataset; unclear if version control is implemented
Singapore	Various formats; CSV/XML uncommon	Almost complete; some datasets lack descriptions or detailed variable definitions	Available	Older data compiled into one dataset; unclear if version control is implemented
European Union	Various formats; CSV/XLS commonly available	Incomplete for some datasets; missing publisher/creator info; minimal license details	Available	Unable to access older dataset versions; past datasets may be compiled into one dataset

The comparison reveals notable differences in data quality across the five portals. Malaysia excels in providing comprehensive metadata and consistently offering CSV and Parquet formats, ensuring users can easily access data. The US and Singapore portals also present clear metadata but vary in file format availability, potentially frustrating users seeking specific formats. The UK portal stands out for its robust version control, enabling users to track dataset updates, while the EU portal struggles with incomplete metadata and limited version control.

CSV availability proves crucial for data accessibility due to its versatility and widespread use in data analysis tools. Keeping (2024) highlights that CSV files are favoured in Big Data and Machine Learning environments for their simplicity as an exchange format. Furthermore, Alpatov and Bogatireva (2024) emphasize CSV's continued relevance as a common format for structured data in analytical systems. The inconsistent presence of CSV across some portals, particularly in the UK and Singapore, may hinder users who rely on this widely adopted format.

Data Findability

Effective data findability plays a crucial role in enabling users to locate relevant datasets efficiently. Key features such as categorization, filtering, and sorting are essential for improving searchability and enhancing user experience. Incorporating rich metadata is vital for this process, as it allows datasets to be organized with meaningful labels, improving accessibility through clear organization and detailed descriptions. Additionally, aligning with established repository best practices ensures that datasets are structured with clear categories and consistent labels, simplifying navigation and supporting improved data discoverability (Wu et al., 2024). By combining these approaches, data platforms can create a seamless search experience that enables users to filter results intuitively and locate relevant datasets with ease.

The following comparison examines how the five data portals implement these principles, focusing on their approaches to categorization, filtering, and sorting to enhance data findability.

Table 2.5 Data Findability Comparison of Malaysia, UK, US, Singapore, and EU Open Data Portals

Country	Effective Categorization and Filtering	Search Functionality	Sorting Options for Datasets
Malaysia	Clear category labels in dataset lists; agency and category filters	Search by title/description	Sort by relevance (date sorting via advanced filters)
United Kingdom	Straightforward filters (publisher, topic, format); chosen filters clearly visible	Basic search options	Sort by best match or most recent
United States	Detailed filters ideal for large repositories; categories shown but not prominent	Basic and advanced search options available	Sort by popularity, relevance, name, or date
Singapore	Useful basic filters (topics, columns, period, formats, agencies); clear dataset labels	Basic search options	Sort by relevance, downloads, or dates
European Union	Basic and advanced filters; may be complex for general users	Search includes advanced features	Sort by popularity, relevance, name, or date

The comparison highlights that while all five portals implement effective data findability strategies, their approaches vary in complexity and user-friendliness. Malaysia's clear category labels and straightforward search functionality make it highly accessible, which is particularly beneficial for introducing a new data hub in a region that is still developing technologically. In contrast, the US and EU portals offer advanced search options and detailed filters, ideal for experienced users but potentially overwhelming for beginners. The UK and Singapore portals strike a balance, providing clear filters and useful sorting options that cater to both novice and experienced users.

For a new data portal, prioritizing essential features such as commonly used filters like agencies and categories/topics, along with a simple search method, ensures users can easily explore available datasets. As the data hub grows and the dataset collection expands, incorporating enhancements like advanced filters or improved sorting options can gradually improve data discoverability without sacrificing ease of use.

Conclusion

Comparing existing government open data portals helped identify key aspects that contribute to an effective data portal. Emphasis on straightforward and user-friendly navigation ensures a seamless browsing experience, preventing users from feeling stuck or lost. Clear and complete dataset metadata plays a crucial role in improving overall data quality, which directly enhances both data understandability and findability. Additionally, features that support dataset exploration, such as a simple search function and commonly used filters like categories and agencies, are essential for guiding users to relevant information efficiently. These insights provide a strong foundation for designing a data portal that balances accessibility with functionality. By prioritizing said aspects, the platform can offer an intuitive experience while laying a strong foundation for future improvements.

3. METHODOLOGY

Requirement Analysis

BayuData was designed to facilitate dataset exploration for public users. To ensure the platform met user Requirements, the following requirements were identified:

User Roles and Requirements

- General Users:
  - Access and explore datasets through search functionality and filtering options.
  - Explore datasets using search functionality and filtering options for categories and agencies, along with basic metadata such as descriptions, dataset titles, and date added.
  - Access dataset details with detailed metadata, including dataset field descriptions, license information, dataset preview functionality, and download options for CSV and Parquet formats.
  - Navigate the platform easily through a clear menu structure and intuitive layout.

- Admins (External Platform):
  - Upload datasets via an external government platform.
  - Review dataset contents to ensure they are appropriate for public sharing and do not contain sensitive or private information before submitting them for approval.

- Supervisor (External Platform):
  - Approve datasets that had been reviewed by admins, ensuring data quality and compliance.

Since the admin and supervisor tasks are performed on an external system beyond the scope of this project, this platform focuses solely on improving dataset accessibility for general users.

Key Features Identified

Based on the analysis of existing open data portals and identified user needs, the following features were prioritized:

- Dataset Exploration**:**
  - Users can browse a dataset list page that includes filters for categories and agencies, along with a search function to refine results efficiently.

- Detailed Dataset View:
    - Users can preview chosen datasets in greater detail, including comprehensive metadata, dataset field descriptions, and download options in CSV and Parquet formats.

- Data Archives:
    - With each dataset update, previous versions are archived and presented in a separate tab, ensuring users have access to historical data.

- Dashboard Exploration (Planned Feature):
    - While the current platform includes a dashboard page layout, the dashboard's content is not yet implemented as it falls outside the project scope. However, this feature is planned for future development.

- User Interface Design:
    - The platform prioritizes simplicity, ensuring users can navigate easily without technical expertise.

By focusing on these aspects, BayuData was tailored to meet the needs of Sabah's public sector and general users while providing a solid foundation for future enhancements.

System Design

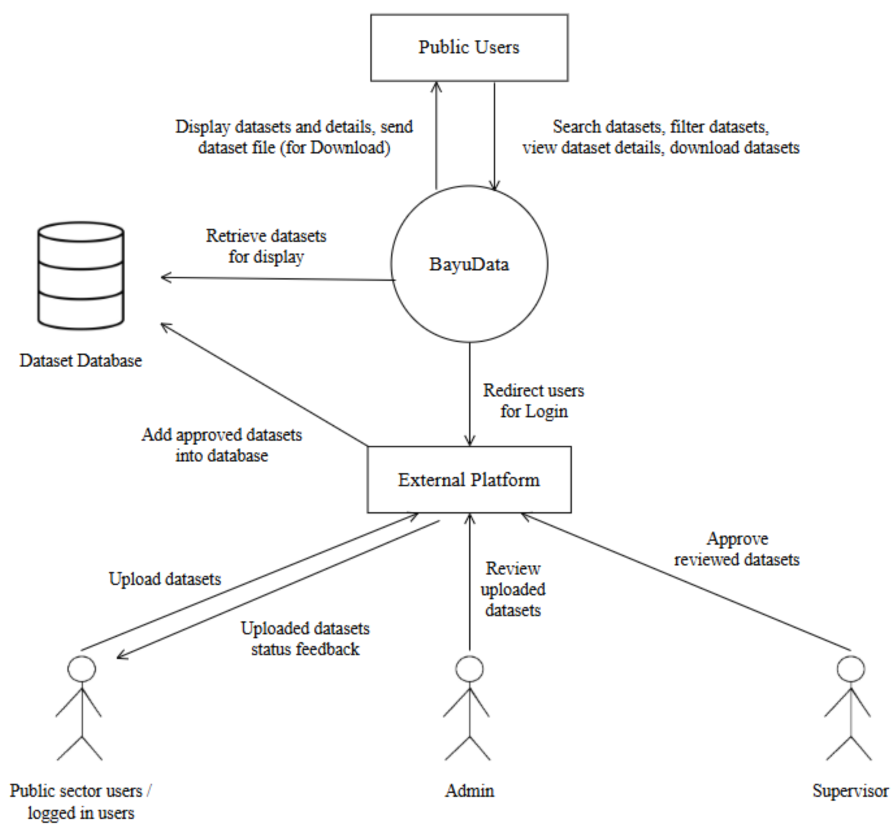

Figure 3.1 System Context Diagram for BayuData

Figure 3.1 illustrates the interaction between BayuData, the external platform, and various user roles. Public users access BayuData to search, filter, and download datasets, while public sector staff users who wish to log in are redirected to the external platform to proceed logging in and upload datasets. Admins review uploaded datasets, and supervisors approve them. Approved datasets are added to the dataset database, from which BayuData retrieves information for display. Users also receive feedback regarding their uploaded dataset status.

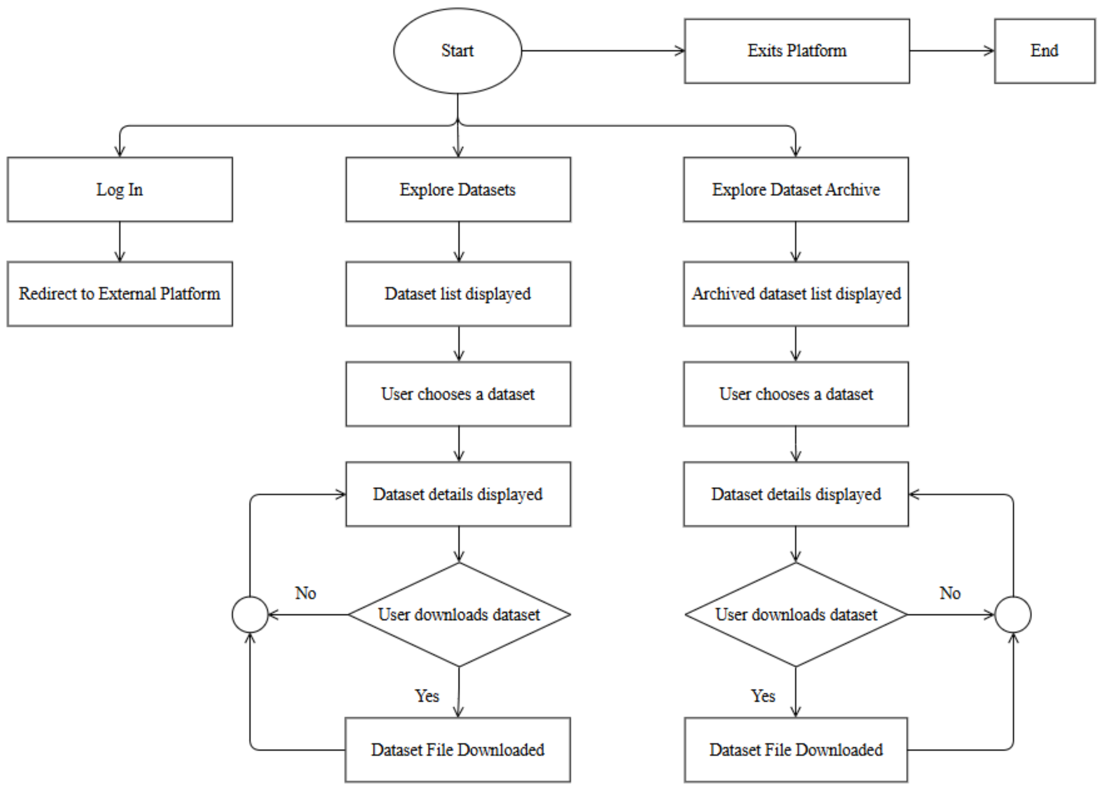

Figure 3.2 System Flowchart for BayuData

Figure 3.2 illustrates the flow of user interactions within BayuData. Users can choose to log in, explore datasets, or explore the dataset archive. Users who choose to log in are redirected to the external platform, where they can upload, review, or approve datasets depending on their user role.

For users exploring datasets or dataset archives, they are presented with a list of available datasets. Upon selecting a dataset, its details are displayed. Users may then choose to download the dataset. If they proceed, the dataset file is downloaded to their device; if not, they can continue looking through the dataset details. The flowchart also includes an option for users to exit the platform at any point, terminating their session.

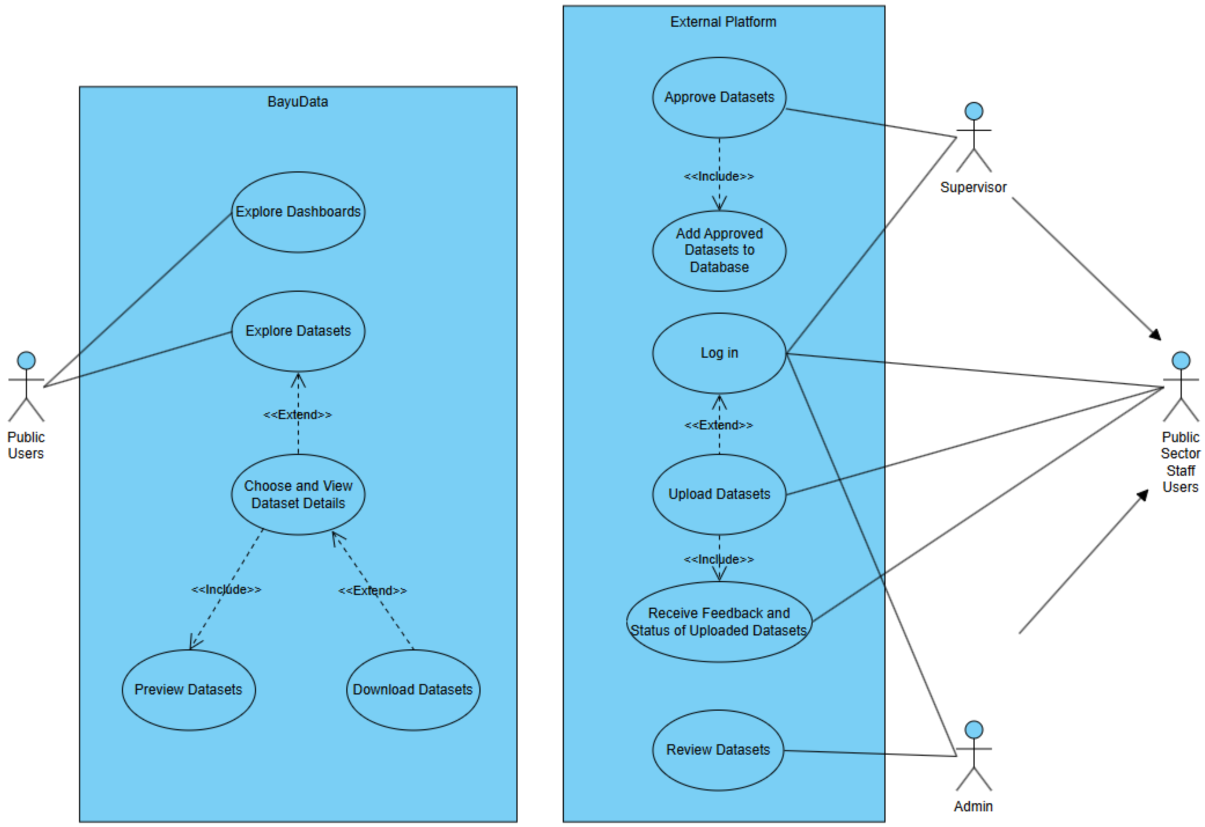

Figure 3.3 Use Case Diagram for BayuData

Figure 3.3 illustrates the interaction between various user roles and system functionalities within BayuData and the associated external platform. Although the external platform is not part of the BayuData system itself, it plays a crucial role in the dataset management process, making its inclusion essential for understanding the overall flow.

In BayuData, public users can either explore dashboards or explore datasets. While the dashboard section currently contains only a placeholder for future content, its inclusion in the diagram signifies planned functionality. When exploring datasets, users are presented with dataset details, which include a dataset preview integrated directly into the details page. Additionally, users have the option to download the dataset if desired.

For dataset submission, review, and approval processes, users must access the external platform. Access to this platform is restricted to public sector staff, who must log in to perform dataset-related tasks. Only authenticated public sector staff members are permitted to upload datasets on behalf of their respective departments. Upon submission, users can receive feedback regarding the status of their uploaded datasets. Admins are responsible for reviewing these submissions, while supervisors oversee the approval process. Once a dataset is approved, it is added to the database and subsequently becomes accessible within BayuData for public users to explore.

User Interface

The user interface design for BayuData was developed with a focus on simplicity, clarity, and ease of navigation, aligning with the preference for a minimalistic design similar to Malaysia's open data portal. Each interface element was designed to ensure users can efficiently explore datasets, access dataset details, and download data as intended. The design emphasizes clear visual hierarchy, ensuring key features such as dataset browsing and filtering are intuitive for users.

<u>Home Page</u>

Figure 3.4 User Interface for BayuData: Home Page (Desktop View)

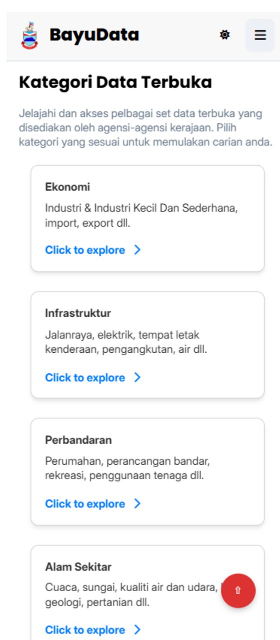

Figure 3.5 User Interface for BayuData: Home Page (Mobile View)

The homepage serves as the landing page for BayuData. It includes a prominent menu bar at the top of the page, providing easy navigation for users. On mobile devices, this menu bar adopts a dropdown format to maintain a clean interface (refer to Appendix A for the mobile view screenshot).

The homepage features a section where users can browse and select from various dataset categories, streamlining access to relevant data. Additionally, there are dedicated buttons that redirect users to the dataset catalogue and the dashboard, ensuring quick access to these key sections.

<u>Data Catalogue</u>

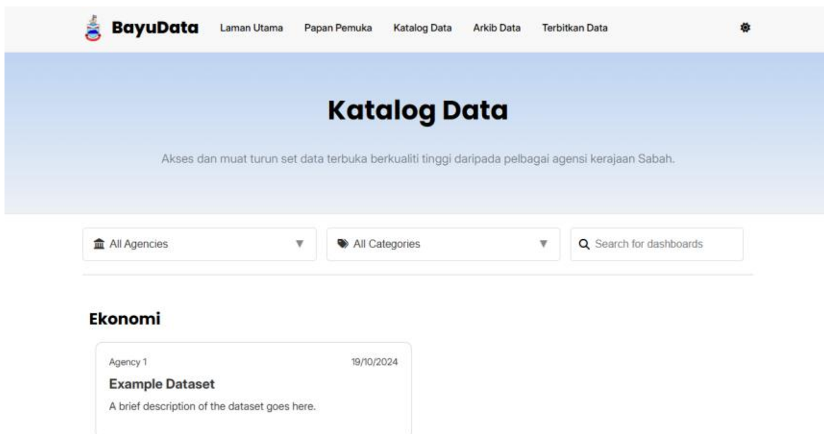

Figure 3.6 User Interface for BayuData: Data Catalogue (Desktop View)

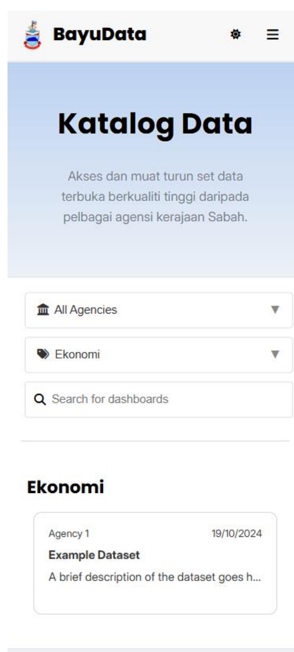

Figure 3.7 User Interface for BayuData : Data Catalogue (Mobile View)

The dataset catalogue displays a list of available datasets, with key information such as dataset titles, description, and date added. Each dataset is also listed under the category label it belongs to. Users can filter datasets by category or agency to refine their search results and utilise the search function. For a detailed view of the search function and the filter function, refer to Appendix B and Appendix C respectively.

Data Archive

The archived dataset catalogue displays older versions of existing datasets. This page follows the same structure and functionality as the dataset catalogue, including dataset details, search, and filter options. Since both catalogues are identical in layout, screenshots have been omitted to avoid redundancy.

Dataset Details

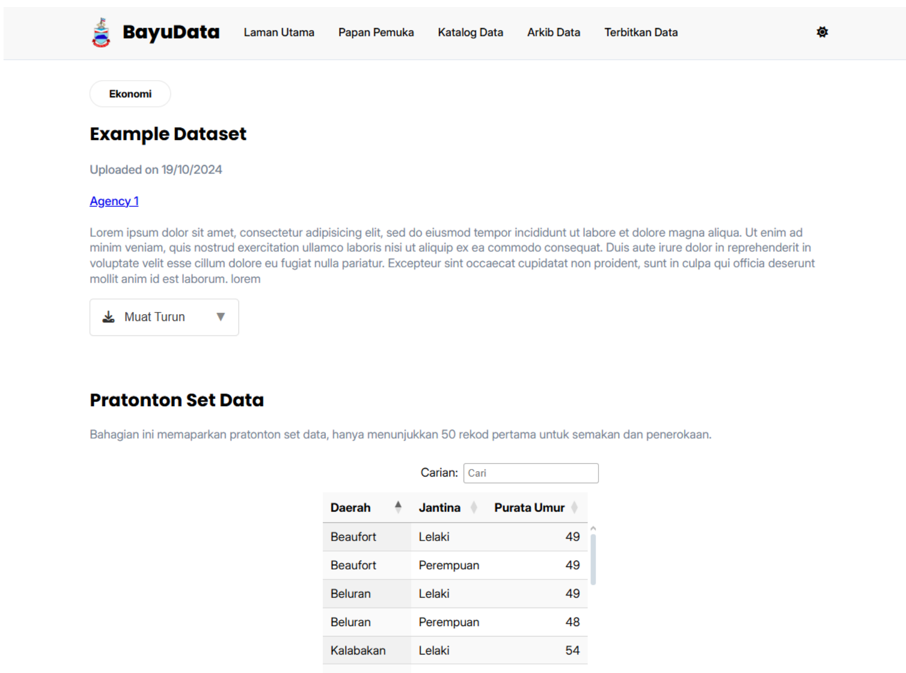

Figure 3.8 User Interface for BayuData : Dataset Details (Desktop View)

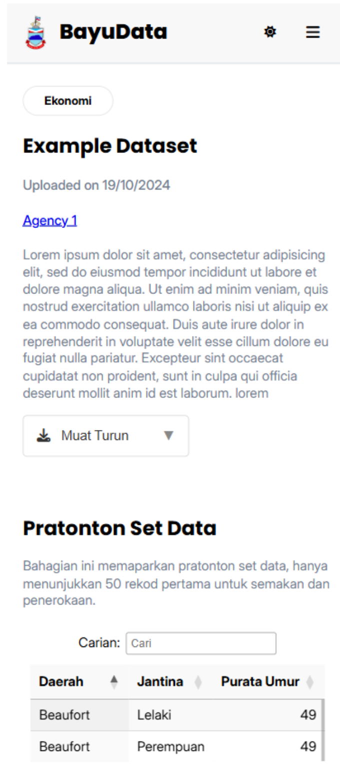

Figure 3.9 User Interface for BayuData : Dataset Details (Mobile View)

The dataset details page provides the user with comprehensive information about the selected dataset. This includes:

- Dataset Preview:
  - Users can preview dataset contents (first 50 rows) directly on this page for quick assessment. Refer to Appendix D for the corresponding screenshot.
- Metadata Information:
  - Key metadata details such as description, variable definition, last update, next scheduled update, data source, dataset URL, and license are displayed for user reference. Refer to Appendix E for the corresponding screenshot.
- Download Section:
  - Users have the option to download the dataset file (either CSV or Parquet) to their device. Refer to Appendix F for the corresponding screenshot.

System Implementation

The design of BayuData's user interface was guided by a deliberate focus on simplicity and minimalism, inspired by Malaysia's open data portal. This decision aimed to provide users with a familiar and intuitive experience while maintaining a clean, user-friendly layout.

At the time of writing, BayuData has not yet been implemented or deployed. While the interface design has been completed, the implementation phase is pending due to certain challenges. One significant obstacle is the lack of appropriate datasets and dashboards required to fully populate the platform and support its intended functionality.

Future plans for implementation include integrating the designed interface with suitable datasets and dashboards once they become available. Additionally, comprehensive testing and evaluation will be conducted to ensure the system performs as intended and meets user requirements.

4. RESULTS AND DISCUSSION

The BayuData project successfully delivered a completed user interface design that aligns with the intended objectives. Emphasizing simplicity and minimalism, the design was inspired by Malaysia's open data portal to ensure users can easily navigate and explore datasets. Key features such as dataset preview, metadata display, and download functionality were effectively integrated to enhance the user experience.

Although the system has not yet been implemented, design reviews conducted with stakeholders provided valuable insights. Feedback indicated that the interface design met expectations in terms of usability, clarity, and alignment with the desired visual style. Stakeholders, particularly my supervisor and fellow team members, responded positively to the design's simplicity and intuitive layout. The decision to maintain a minimalistic design aligned well with the company's preference for a clean and user-friendly interface. This design approach avoids overwhelming users with cluttered elements and ensures their focus remains on essential content rather than being distracted by strong colour options.

The project also contributed to the development of valuable skills, particularly in web development, responsive web design, and the utilization of existing JavaScript extensions such as DataTables. Aligning design decisions with user requirements while ensuring consistency with established standards was crucial. Additionally, effective communication with stakeholders and adapting to their recommendations played a key role in the design process.

The completed design provides a structured foundation for the future implementation of BayuData. By adopting a familiar and user-friendly interface, the design aims to improve data accessibility, streamline dataset exploration, and enhance overall user satisfaction. This aligns with the company's goal of promoting open data engagement among the general public and public sector staff. The project also represents a step forward in improving transparency between the people of Sabah and the government, contributing to Sabah's progress in adopting technological advancements.

To further improve BayuData, several enhancements can be considered in future updates. This includes incorporating visual elements for dashboards, refining the dataset preview layout for improved clarity, and enhancing metadata presentation to provide users with more comprehensive information. Additional improvements such as language options, APIs, user documentation, and interactive data exploration tools can also be introduced to further align BayuData with its objective of facilitating efficient data discovery and exploration.

5. CONCLUSION

The BayuData project aimed to develop a user interface design that simplifies data exploration for users while maintaining a clean and minimalistic layout. Inspired by Malaysia's open data portal, the design was tailored to meet stakeholder preferences for simplicity and ease of navigation. Key features such as dataset previews, metadata display, and download options were effectively incorporated to enhance functionality.

Although the system is yet to be implemented, the completed design establishes a solid foundation for future development. Stakeholder feedback affirmed that the design meets expectations in terms of usability, visual style, and alignment with the project's goals. The decision to prioritize a straightforward layout effectively addresses the company's objective of improving data accessibility for public sector staff and the public.

The project also contributed to enhancing skills in web design, responsive layout implementation, and effective communication with stakeholders. Overcoming design challenges and aligning with organizational preferences strengthened the overall design outcome.

Moving forward, the BayuData design offers a practical solution that supports open data engagement. With its intuitive interface, the design is well-positioned to improve public access to government datasets, ultimately contributing to greater transparency and informed decision-making within the community.

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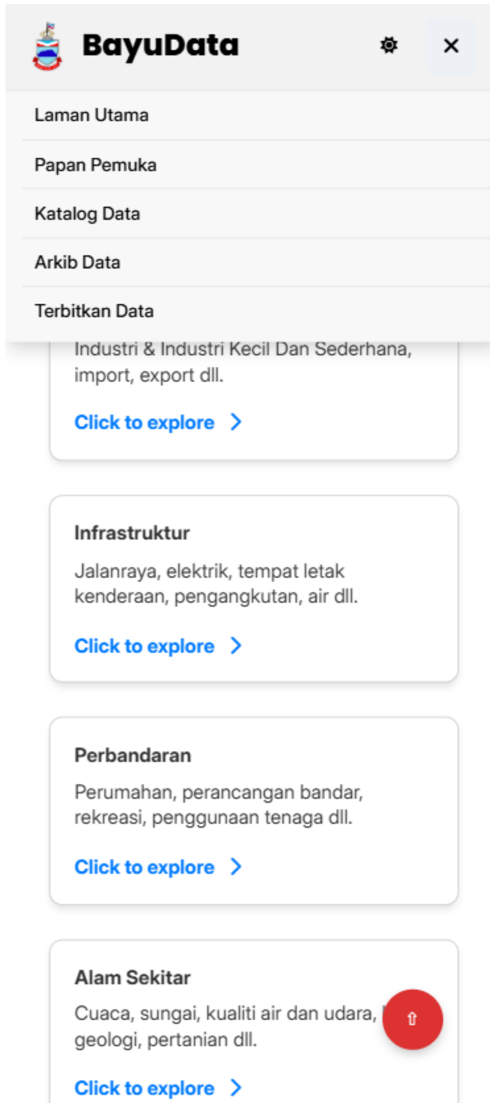
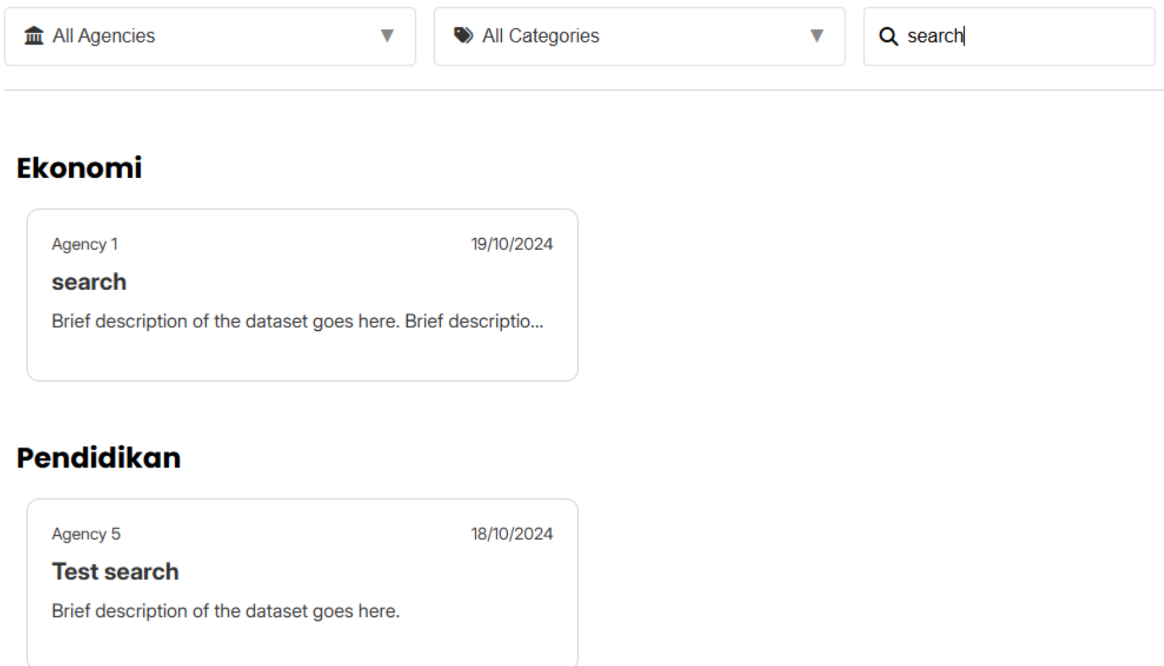
APPENDIX A

Screenshot of the Drop-Down Menu in Mobile View



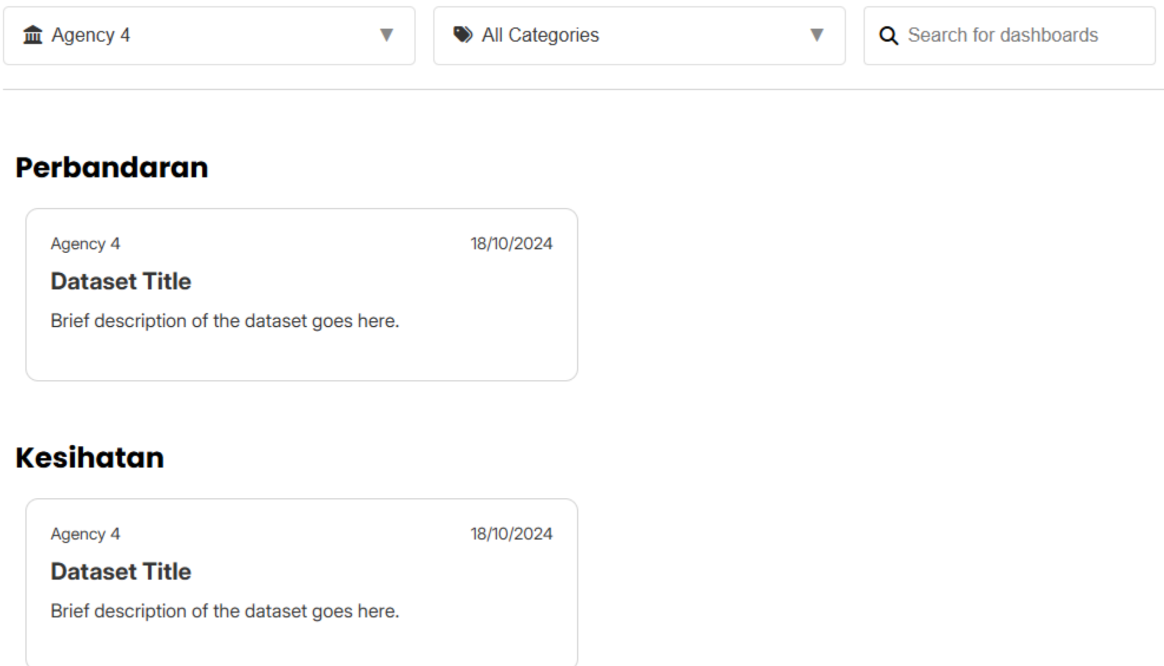
APPENDIX B

Screenshot of the Search Function in Data Catalogue and Archived Data Catalogue



APPENDIX C

Screenshot of the Filter Function in Data Catalogue and Archived Data Catalogue



APPENDIX D

Screenshot of the Data Preview Section in Dataset Details Page

Pratonton Set Data

Bahagian ini memaparkan pratonton set data, hanya menunjukkan 50 rekod pertama untuk semakan dan penerokaan.

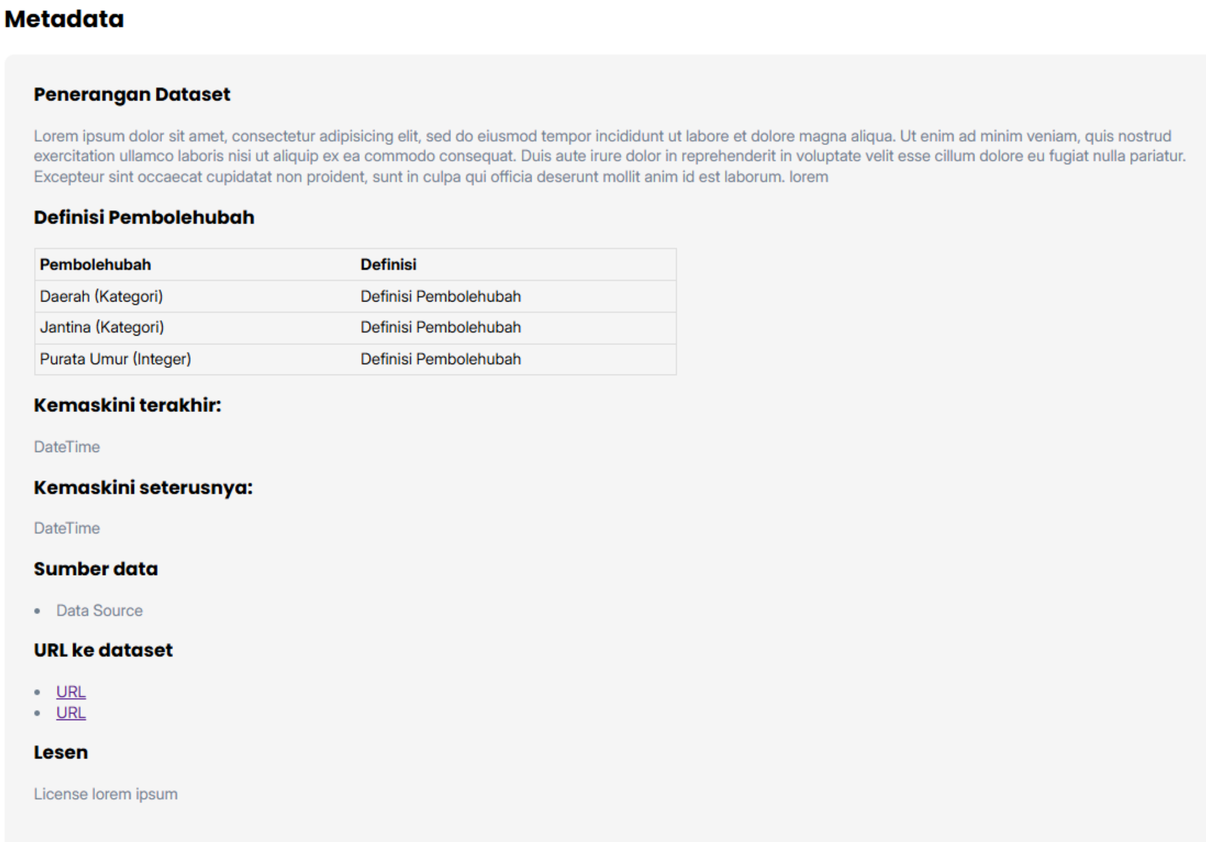
Carian: Cari

Daerah	Jantina	Purata Umur
Kuala Penyu	Lelaki	47
Kuala Penyu	Perempuan	45
Kudat	Lelaki	52
Kudat	Perempuan	52
Kunak	Lelaki	54
Kunak	Perempuan	46
Lahad Datu	Lelaki	48
Lahad Datu	Perempuan	47
Membakut	Lelaki	48
Membakut	Perempuan	46

Paparan dari 1 hingga 50 dari 50 rekod

APPENDIX E

Screenshot of the Metadata Section in Dataset Details Page



Metadata

Penerangan Dataset

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. lorem

Definisi Pembolehubah

Pembolehubah	Definisi
Daerah (Kategori)	Definisi Pembolehubah
Jantina (Kategori)	Definisi Pembolehubah
Purata Umur (Integer)	Definisi Pembolehubah

Kemaskini terakhir:

DateTime

Kemaskini seterusnya:

DateTime

Sumber data

- Data Source

URL ke dataset

- URL
- URL

Lesen

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APPENDIX F

Screenshot of the Download Section in Dataset Details Page